

VINAYKUMAR B PATIL

EMBEDDED SYSTEMS ENGINEER

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SUMMARY

Embedded Systems Engineer with 2+ years of experience developing production firmware for aerospace, UAV, and power-electronics applications. Proficient in C/C++, STM32 HAL/LL, and FreeRTOS-based preemptive multitasking systems. Experienced in multi-protocol driver development (UART, SPI, I2C, CAN), automated unit and regression testing, board bring-up, and hardware-software integration. Comfortable in cross-functional environments with defect tracking, SCM tools, and peer-review workflows.

TECHNICAL SKILLS

Languages	C, Embedded C, C++ (primary); C# (exposure)
RTOS	FreeRTOS (preemptive multitasking); QNX RTOS (familiar)
Microcontrollers	STM32 (HAL/LL), ESP32, Arduino
Protocols	UART, SPI, I2C, CAN, SBUS, PPM
Testing	Automated unit testing, regression testing, hardware-in-loop validation
Hardware	PCB Design (Altium, KiCad), board bring-up, logic analyzer, CRO, multimeter
Tools	STM32CubeIDE, Git, Jira (defect tracking), peer code review

EXPERIENCE & PROJECTS

Flying Wedge Defence & Aerospace

Feb 2024 – Present

Embedded Systems Engineer

- Designed and implemented STM32 (HAL/LL) firmware for avionics and sensor subsystems on UAV platforms, operating in preemptive multitasking real-time environments using FreeRTOS.
- Developed and validated UART, SPI, I2C, and CAN communication drivers; reduced integration defects through structured peer code reviews and hardware-in-loop testing.
- Built automated unit and regression test suites for C/C++ firmware modules, enabling repeatable validation across firmware releases.
- Implemented digital calibration algorithms for precision current and voltage monitoring, improving measurement accuracy across production hardware.
- Designed and validated sensing and power-monitoring PCBs (Altium/KiCad); performed full board bring-up and hardware debugging for production prototypes.
- Assessed open-source MAVLink libraries for C2 suitability; integrated GPS, IMU, and RF peripherals into the embedded software stack.
- 2.4 GHz Wireless Transceiver for UAV Control* — Designed transmitter-receiver system supporting 10 PWM channels with SBUS/PPM output; implemented SPI and I2C drivers for RF modules and peripheral communication.
- Precision Current & Voltage Monitoring Module* — Designed high-accuracy measurement PCB with digital calibration firmware; validated output against traceable references.

Bosch Global Software Technologies

Jan 2023 – May 2023

Software Development Engineer Intern

- Developed embedded communication firmware for camera-based automotive systems; implemented real-time actuator control logic improving response latency and system reliability.
- Participated in defect tracking and resolution using internal SCM tools; performed peer code reviews following automotive firmware coding standards.
- Aeronero Air-to-Water System (IoT Embedded Platform)* — Integrated multi-sensor data acquisition, real-time calibration, and Wi-Fi telemetry on ESP32 for a cloud-connected atmospheric water generation platform.

EDUCATION

B.E. Electronics and Communication Engineering

2019 – 2023

KLE Technological University, Hubballi